RESTRICTION PROPERTY TRANSFER ALONG A DOUBLE RESTRICTION

THE STACKS PROJECT AUTHORS

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Editorial note. Manual completeness read applied: the proof delegates only to the statement-level commutative cartesian diagram of lemma-double-restrict written in this chapter, and writes out both case-specific observations (image containment; pr_0 as base changed covering); the remaining steps are formal single-step diagram consequences, so the proof is judged complete under the same criterion applied to L22533.

Allowed standard prerequisites (editorial).

- The double restriction diagram: h, h', h" with a commutative diagram whose both squares are cartesian (more-groupoids chapter, Lemma lemma-double-restrict, canonical tag 04MW; same-chapter item, triaged out of this lane's pool for its own proof but statement-level diagram fully written). cited at proof lines 1105-1106. Supplies the commutative and cartesian squares from which both directions follow formally.

Source reference. Original lemma 04MW: sources/stacks-chow/tags.txt, source lines 6057–6057. Cited at proof lines 1105-1106: both directions follow formally from the commutative diagram with both squares cartesian written in its statement.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Let $g : U' \to U$ and $g' : U'' \to U'$ be morphisms of schemes. Set $g'' = g \circ g'$. Let (U', R', s', t', c') be the restriction of R to U'. Let $h = s \circ pr_1 : U' \times_{g,U,t} R \to U$, let $h' = s' \circ pr_1 : U'' \times_{g',U',t} R \to U'$, and let $h'' = s \circ pr_1 : U'' \times_{g'',U,t} R \to U$. Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf, fpqc\}$. Let $\mathcal{P}$ be a property of morphisms of schemes which is preserved under base change, and which is local on the target for the τ-topology. If*

(1) *$h(U' \times_U R)$ is open in U,*
(2) *$\{h : U' \times_U R \to h(U' \times_U R)\}$ is a τ-covering,*
(3) *h' has property $\mathcal{P}$,*

then h'' has property $\mathcal{P}$. Conversely, if

(a) *$\{t : R \to U\}$ is a τ-covering,*

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(d) *h'' has property $\mathcal{P}$,*

then h' has property $\mathcal{P}$.

Proof. This follows formally from the properties of the diagram of Lemma 04MW. In the first case, note that the image of the morphism h'' is contained in the image of h, as $g'' = g \circ g'$. Hence we may replace the U in the lower right corner of the diagram by $h(U' \times_U R)$. This explains the significance of conditions (1) and (2) in the lemma. In the second case, note that $\{\mathrm{pr}_0 : U' \times_{g,U,t} R \to U'\}$ is a τ-covering as a base change of τ and condition (a). $\square$